

Supplementary Materials

Stream Reduction in Multi-Stream Graph Convolutional Networks: A Non-Inferiority Study for Skeleton-Based Action Recognition

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S1. Numerical Evidence and Reproduction Scope

The final plan comprised four trainings, four stream evaluations, and one paired analysis. The source audit verified all nine receipts and their 31 output files. Four final checkpoints contained epoch 65, the planned configurations and bindings, and finite tensor values. A separately implemented analysis reproduced the fixed float32 logit fusion, predicted classes, confusion matrices, and all 10,000 three-endpoint bootstrap replicates exactly. The separate implementation aggregates per-setup true, predicted, and true-positive counts; the original implementation uses weighted confusion matrices.

These checks apply to the selected final evidence. The transported selection covered 370 of 614 entries in the original manifest. The remaining 244 entries concern intermediate epoch or evaluation-chunk tensors. This is not a claim that the original TAR was directly checked in its entirety. The public curation preserves the outputs needed to reproduce the final numerical comparison; it does not include raw NTU skeleton arrays or a certified portable raw-data-to-trained-model workflow.

The earlier latency experiment provides 18 paired traces over three folds, two mapping regimes, and three process repetitions. Saved traces support CPU reanalysis of timings, not a new measurement of final-model latency. The original final training environment was Python 3.11.16, PyTorch 2.11.0 with CUDA 12.8, and NumPy 2.4.6. Separate saved-output analysis used Python 3.12.13 and NumPy 2.3.5. Exact floating-point agreement was observed in those recorded analyses; identical bits on arbitrary future environments are not guaranteed.

S2. Training and Temporal Representation

All final streams use the configurations reported in the main text: 65 epochs, final-epoch selection without validation monitoring, and separate fixed stream seeds. The sparse adapter reads interpolation supports, then reconstructs 64 temporal outputs for K64. Nominal temporal indices saved in the prediction files describe output positions and should not be mistaken for all source supports read by interpolation. A metadata and tensor parity check on 432 base-training cases preceded final execution; it did not use test tensors. The original scientific source and immutable configurations are retained in the research package.

S3. Complete Setup Results (Table S1)

Accuracy values are percentages; differences are percentage points. A61 n denotes the number of A61–A120 samples. Zero support is not an observed zero accuracy. Lost and gained count paired correct-to-error and error-to-correct transitions, respectively. All entries are descriptive.

Setup	n	A61 n	Full-4	Stream-3	Difference	Lost	Gained
S01	2872	0	91.191	91.295	+0.104	20	23
S03	3234	0	93.878	93.754	-0.124	29	25
S05	2866	0	91.382	91.382	+0.000	22	22
S07	4288	0	91.325	91.465	+0.140	27	33
S09	2510	0	94.422	94.542	+0.120	8	11
S11	4662	0	91.098	90.819	-0.279	49	36
S13	3947	0	88.751	88.954	+0.203	35	43

Setup	n	A61 n	Full-4	Stream-3	Difference	Lost	Gained
S15	2854	0	90.434	90.399	-0.035	24	23
S17	2869	0	88.184	88.567	+0.383	29	40
S19	3224	3224	80.676	80.428	-0.248	41	33
S21	2144	2144	85.961	85.354	-0.606	34	21
S23	5032	5032	88.434	88.235	-0.199	59	49
S25	2517	2517	84.545	84.783	+0.238	27	33
S27	5384	5384	87.760	87.296	-0.464	84	59
S29	5343	5343	87.367	87.273	-0.094	64	59
S31	5731	5731	90.054	89.984	-0.070	53	49

S4. Complete Action Results (Table S2)

Every action is retained in action-ID order. Recall and F1 are percentages; recall differences are percentage points. These are descriptive summaries, not 120 individually confirmed non-inferiority claims. F1 also depends on predicted positives, so its direction of change can differ from that of recall.

Action	n	Recall Full-4	Recall Stream-3	Difference	F1 Full-4	F1 Stream-3
A001	500	92.400	92.000	-0.400	92.771	92.555
A002	500	85.800	85.800	+0.000	86.405	85.972
A003	499	90.782	90.581	-0.200	89.614	88.889
A004	500	91.200	90.800	-0.400	92.495	91.440
A005	502	91.434	91.633	+0.199	92.540	92.462
A006	502	97.410	97.211	-0.199	96.832	97.114
A007	502	89.841	88.845	-0.996	91.574	90.650
A008	497	97.988	97.787	-0.201	98.783	98.381
A009	498	98.594	98.795	+0.201	99.192	99.094
A010	500	89.200	89.400	+0.200	88.404	87.733
A011	500	64.200	65.000	+0.800	63.881	64.166
A012	498	63.655	63.253	-0.402	56.506	57.117
A013	499	90.381	89.980	-0.401	90.927	89.890
A014	502	99.602	99.402	-0.199	96.246	95.138
A015	504	97.619	98.413	+0.794	97.813	97.927
A016	502	89.841	90.040	+0.199	86.069	86.840
A017	502	81.673	83.068	+1.394	83.080	83.483
A018	504	93.254	92.857	-0.397	91.977	91.585
A019	504	92.063	92.262	+0.198	92.155	92.170
A020	502	97.610	97.410	-0.199	97.222	97.024
A021	502	97.012	97.012	+0.000	97.012	96.915
A022	503	98.211	98.211	+0.000	93.561	92.683
A023	504	93.056	93.056	+0.000	95.132	94.461
A024	503	94.433	94.235	-0.199	94.810	94.611
A025	504	93.254	93.056	-0.198	91.977	90.106
A026	504	99.206	99.206	+0.000	98.717	98.814
A027	504	99.206	99.206	+0.000	99.108	99.305
A028	501	88.623	88.024	-0.599	90.890	90.741
A029	502	74.900	75.100	+0.199	73.223	72.850
A030	502	79.084	82.271	+3.187	72.182	71.889
A031	501	87.026	87.425	+0.399	86.594	86.051
A032	504	91.071	90.873	-0.198	92.354	92.525
A033	503	90.855	90.656	-0.199	91.583	91.200
A034	503	88.072	88.668	+0.596	87.723	87.365
A035	504	99.603	99.405	-0.198	98.335	98.332
A036	503	96.819	96.620	-0.199	94.471	93.732
A037	502	90.837	90.438	-0.398	87.777	86.724
A038	504	91.468	92.659	+1.190	90.659	91.033
A039	504	93.254	93.254	+0.000	93.719	93.812
A040	498	96.988	96.386	-0.602	96.794	96.774
A041	504	85.317	84.921	-0.397	84.980	84.921

Action	n	Recall Full-4	Recall Stream-3	Difference	F1 Full-4	F1 Stream-3
A042	503	99.404	99.602	+0.199	98.232	98.139
A043	504	96.032	96.032	+0.000	97.877	97.877
A044	503	80.119	81.113	+0.994	84.664	84.823
A045	504	85.714	84.921	-0.794	89.441	89.074
A046	504	89.881	88.889	-0.992	92.733	91.709
A047	504	87.500	86.310	-1.190	89.725	89.139
A048	503	87.873	89.066	+1.193	90.020	90.964
A049	503	94.235	92.843	-1.392	95.181	93.775
A050	499	88.978	88.978	+0.000	91.358	91.358
A051	502	91.235	90.438	-0.797	93.469	92.938
A052	503	97.217	97.614	+0.398	97.313	97.421
A053	504	83.929	84.127	+0.198	86.948	87.064
A054	497	92.354	93.964	+1.610	86.767	88.196
A055	480	97.083	96.875	-0.208	97.185	97.077
A056	504	91.468	91.468	+0.000	90.837	91.197
A057	503	88.270	88.867	+0.596	90.520	90.486
A058	504	96.032	95.833	-0.198	97.384	97.183
A059	499	99.800	100.000	+0.200	97.647	97.843
A060	503	95.626	96.024	+0.398	95.721	96.311
A061	479	93.319	92.902	-0.418	94.603	94.080
A062	484	94.628	93.802	-0.826	90.783	92.183
A063	487	93.634	94.251	+0.616	89.852	88.953
A064	489	96.319	96.115	-0.204	97.314	97.208
A065	488	91.393	90.779	-0.615	93.014	92.484
A066	490	96.122	96.122	+0.000	96.615	96.122
A067	488	77.459	78.074	+0.615	77.459	76.892
A068	490	87.143	85.306	-1.837	89.331	88.093
A069	490	84.286	84.286	+0.000	81.379	81.944
A070	490	89.184	89.796	+0.612	88.731	88.710
A071	490	67.959	67.959	+0.000	71.845	72.234
A072	490	70.408	71.224	+0.816	64.126	63.919
A073	488	57.172	54.713	-2.459	56.707	56.034
A074	486	65.844	65.844	+0.000	71.588	71.829
A075	481	74.012	73.389	-0.624	70.286	71.748
A076	488	52.254	54.508	+2.254	57.562	59.574
A077	489	76.483	76.074	-0.409	81.838	81.311
A078	488	72.541	72.336	-0.205	77.974	78.707
A079	488	86.680	85.656	-1.025	81.581	81.008
A080	490	98.571	98.367	-0.204	96.697	96.690
A081	489	91.207	91.207	+0.000	92.149	92.340
A082	491	81.874	81.059	-0.815	80.885	79.920
A083	491	81.874	81.059	-0.815	85.532	85.868
A084	489	69.121	69.325	+0.204	72.688	73.218
A085	489	89.775	88.957	-0.818	89.137	88.866
A086	489	87.526	87.117	-0.409	85.858	85.371
A087	491	94.501	94.501	+0.000	95.769	96.066
A088	492	96.545	96.341	-0.203	96.154	96.244
A089	491	84.521	82.892	-1.629	87.924	86.412
A090	492	93.293	93.089	-0.203	90.891	90.424
A091	490	82.653	81.633	-1.020	82.737	82.136
A092	482	95.021	94.398	-0.622	94.726	94.301
A093	490	86.939	86.531	-0.408	86.585	87.603
A094	487	88.296	87.680	-0.616	89.397	89.237
A095	489	97.137	96.524	-0.613	92.323	92.098
A096	490	95.918	96.122	+0.204	95.528	95.441
A097	491	98.371	98.167	-0.204	99.077	98.872
A098	490	98.776	98.776	+0.000	98.374	98.474
A099	491	96.130	96.741	+0.611	98.027	98.242
A100	491	98.371	98.574	+0.204	98.371	98.374
A101	491	99.593	99.593	+0.000	97.898	98.291
A102	491	95.927	96.538	+0.611	95.538	95.758
A103	492	72.764	73.374	+0.610	74.043	75.681
A104	492	89.837	89.431	-0.407	93.943	93.617

Action	n	Recall Full-4	Recall Stream-3	Difference	F1 Full-4	F1 Stream-3
A105	491	76.986	75.153	-1.833	72.903	72.638
A106	491	77.597	76.578	-1.018	78.314	78.415
A107	492	69.715	72.154	+2.439	73.763	74.423
A108	491	91.039	90.835	-0.204	86.880	87.024
A109	492	91.057	90.650	-0.407	91.616	92.244
A110	490	81.633	81.837	+0.204	80.645	81.421
A111	491	92.261	91.650	-0.611	91.515	91.185
A112	492	97.561	97.561	+0.000	98.260	97.959
A113	492	97.764	97.561	-0.203	98.566	98.664
A114	491	94.705	94.705	+0.000	96.174	95.876
A115	491	91.650	91.650	+0.000	89.197	90.090
A116	492	94.715	95.528	+0.813	95.688	96.213
A117	491	87.373	87.576	+0.204	90.506	90.526
A118	492	95.122	94.309	-0.813	94.545	94.213
A119	490	95.510	95.510	+0.000	94.929	94.833
A120	492	97.358	96.951	-0.407	97.259	97.446

S5. Descriptive Paired Decision Changes

This extension was computed from the already audited paired result file after the final test comparison. It reads the saved fused predictions and verifies their consistency with saved logits and confusion matrices. It performs no training, new test forward, refitting, or additional hypothesis test. The fusion weights, non-inferiority margins and original bootstrap draws are unchanged.

Table S3: Correctness changes by true action group. Same and different refer to the two predicted labels among examples misclassified by both systems.

Group	n	Both correct	Lost	Gained	Wrong, same	Wrong, different
All 120	59,477	52,392	605	559	5,307	614
A1–A60	30,102	27,178	243	256	2,215	210
A61–A120	29,375	25,214	362	303	3,092	404

The complete outputs preserve all 120 classes and all 120×120 label pairs, including zero counts. The directed transition matrix records the Full-4 predicted label as the source and the Stream-3 predicted label as the destination. A separate true-class/error-class table counts introduced and corrected errors against the same erroneous alternative. These are distinct representations and must not be interchanged.

Table S4 lists the ten most frequent off-diagonal prediction transitions, ordered by decreasing count, then ascending source and destination action IDs to resolve ties. The ordering was determined descriptively after inspecting the final predictions. It does not define an independently confirmed subgroup effect.

Table S4: Most frequent directed changes in predicted label. Lost: Full-4 correct and Stream-3 incorrect. Gained: Full-4 incorrect and Stream-3 correct. Both wrong: both predictions incorrect.

Full-4 label	Stream-3 label	Lost	Gained	Both wrong	Total
A073	A076	14	18	1	33
A012	A030	6	14	12	32
A071	A072	7	14	6	27
A072	A071	8	10	6	24
A016	A017	3	10	3	16
A106	A107	8	5	3	16
A069	A072	7	2	6	15
A011	A012	4	4	6	14
A075	A073	3	6	5	14
A074	A084	6	5	2	13

For example, A073 to A076 comprises 14 introduced errors on true A073 examples, 18 corrected errors on true A076 examples, and one example with a different true class that both systems misclassified. The official dataset list names A073 “staple book” and A076 “cutting paper” (<https://rose1.ntu.edu.sg/dataset/actionRecognition/>, accessed 20 September 2026). This observation identifies where decisions changed; it does not establish an anatomical or graph-topological cause.

The script `describe_paired_errors.py`, input SHA-256, full CSV matrices, and its environment/provenance record accompany the research artifact. Reported inference remains the original three-endpoint paired setup-bootstrap analysis.